1(B): TE_{10} VERSUS TM_{11}

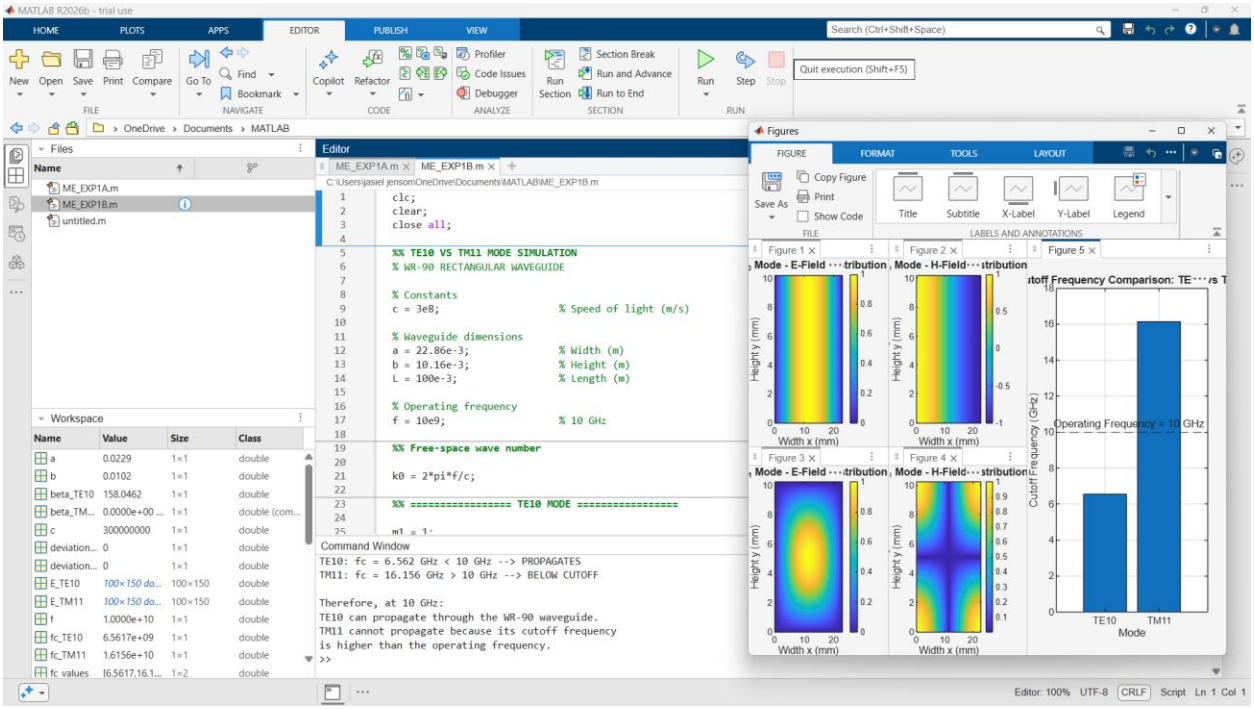
Aim:

To compare TE_{10} and TM_{11} modes in a WR-90 rectangular waveguide by analyzing their cutoff frequencies, propagation characteristics, and field distributions at 10 GHz.

Procedure:

1. Define the WR-90 waveguide dimensions and operating frequency.
2. Calculate the cutoff frequencies for TE_{10} and TM_{11} modes.
3. Determine the propagation constant or attenuation constant for each mode.
4. Plot and compare the electric- and magnetic-field distributions using MATLAB.
5. Identify which modes propagate at 10 GHz.

Output:



Result:

The TE_{10} mode propagates at 10 GHz because its cutoff frequency is approximately 6.56 GHz. The TM_{11} mode has a cutoff frequency of approximately 16.42 GHz and therefore remains below cutoff.